

Calculus 1m - 104018 - Spring Semester

Final Exam A

Instructions:

You must answer on these pages. You may not add pages or papers to this exam.
You must write with a black or a blue pen. Do not use a pencil or a red pen.

You may not use notes, books, calculators or electronic devices of any kind.

Good Luck!

Part A

Mark the correct answer in each of the questions. Each of the first 4 questions is 8 points.

1. Let $f(x) = \begin{cases} \sin \frac{1}{x} & x \neq 0 \\ -\frac{1}{3} & x = 0 \end{cases}$. Mark the correct answer.

- (a) $\int_{-1}^1 f(x)dx = 0$.
- (b) The function is one to one (injective) on $(0, \infty)$.
- (c) $\int_0^1 f(x)dx > 1$.
- (d) $f(x)$ is increasing on $(-\infty, 0)$.
- (e) $f'(x)$ is bounded on the interval $(0, \infty)$.

2. Consider the improper integrals

$$I = \int_1^{\infty} x \sin\left(\frac{1}{x^{\frac{1}{2}}}\right) \arctan\left(\frac{1}{x}\right) dx \quad II = \int_0^1 \frac{\ln x}{\sin(\sqrt{x})} dx.$$

- (a) I does not converge and II converges.
- (b) Both improper integrals converge.
- (c) I converges and II does not converge.
- (d) None of the improper integrals converges.

3. Consider the series

$$I = \sum_{n=1}^{\infty} n^2 \left(1 - \cos\left(\frac{1}{n^2}\right)\right) \quad II = \sum_{n=1}^{\infty} \frac{(n+1)^n}{2^n \cdot n!}$$

- (a) Both series converge.
- (b) Both series do not converge.
- (c) Series I does not converge and series II converges.
- (d) Series I converges and series II does not converge.

4. Let $f(x)$ be defined for all x . Mark the correct statement.

- (a) If $f(x)$ is not continuous at 0, then $(f(x))^2$ is not differentiable at 0.
- (b) If $f'_-(x)$, $f'_+(x)$ exist for all x , then $f(x)$ is continuous at every point.
- (c) If $f(x)$ is convex on the interval $(-1, 1)$ then $f(x)$ is differentiable on the interval $(-1, 1)$.
- (d) If $f''(x) > 0$ for all x , then $f(x)$ has a unique minimum point.

5. (12 points) Answer each of the following True or False questions:

(a) If $\lim_{n \rightarrow \infty} a_n$ exists and isn't zero, and $\lim_{n \rightarrow \infty} a_n \cdot b_n$ exists, then $\lim_{n \rightarrow \infty} b_n$ exists.

True False

(b) If $\sum_{n=1}^{\infty} a_n^3$ converges, then $\sum_{n=1}^{\infty} a_n^6$ converges.

True False

(c) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} \sin(a_n)$ converges.

True False

(d) If $a_n \xrightarrow[n \rightarrow \infty]{} 0^+$ (meaning the sequence is positive and tends to zero), then $\ln(1 + a_n) \leq a_n^2$ for all $n \geq 1$.

True False

Part B

Answer each of the following questions fully. Explain all your answers.

6. (28 points)

(a) (5 points) State Rolle's theorem.

(b) (12 points) Prove Rolle's theorem.

- (c) (11 points) Let $f(x)$ be a differentiable function on the interval $[a, b]$ for which $f'_+(a) > 0$, $f'_-(b) < 0$. Prove that there is $a < c < b$ for which $f'(c) = 0$.

7.(a) (9 points) Calculate

$$\int \frac{6x + 7}{x^2 + 4x + 7} dx$$

(b) (10 points) Calculate $\sqrt{1.1} = (1.1)^{\frac{1}{2}}$ with accuracy 10^{-4} .

(c) (9 points) Calculate

$$\lim_{x \rightarrow 0} \frac{\int_{x^2}^{x^3} \sin(t^2) dt}{\sin x^6}$$